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Camera module and terminal device

Abstract

A camera module and a terminal device having the camera module, the camera module including a plurality of lenses, where lenses of the plurality of lenses are sequentially arranged from an object side to an image side along a direction of an optical axis, where at least one of the plurality of lenses is a free-form lens, where the free-form lens is a non-rotationally symmetric lens, where a first lens of the plurality of lenses is a lens of the plurality of lenses nearest the object side in a direction from the object side to the image side, where a distance on the optical axis between an object-side surface of the first lens and an imaging surface is TTL, where an effective focal length of the camera module is EFL, and $TTL/EFL \leq 2.0$.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a national stage of International Application No. PCT/CN2020/106958, filed on Aug. 5, 2020, which claims priority to Chinese Patent Application No. 201911205850.4, filed on Nov. 29, 2019 and Chinese Patent Application No. 201910734247.9, filed on Aug. 9, 2019. All of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(2) This application belongs to the field of optical imaging technologies, and in particular, to a camera module and a terminal device of a photographic optical system.

BACKGROUND

(3) With the popularity of mobile terminal devices, photography technologies for users to obtain images have developed rapidly. To meet a wide range of market needs, large sensor sizes, large apertures, a diversity of fields of view, and compact structures are important development trends for camera modules. From single-camera modules, to dual-camera modules, and to multi-camera modules, a combination of a plurality of field-of-view lenses has been implemented. This also allows ultra wide-angle lens sets to be applied to mobile terminal devices.

(4) Ultra wide-angle lenses have larger fields of view, but it is difficult to balance a contradiction between an optical distortion and a compact structure. Therefore, most ultra wide-angle lenses have relatively large optical distortions, and a TV distortion of an image is also very obvious. In the field of mobile electronic camera modules limited by compact structures, an optical distortion problem is more prominent and is difficult to resolve. In addition, in application of terminal devices, during video recording by using an ultra wide-angle lens, real-time distortion correction for a video image consumes a large quantity of processing resources and is difficult to achieve.

(5) How to resolve the optical distortion problem of the ultra wide-angle lens set should be a direction of research and development in the industry. SUMMARY

(6) Embodiments of this application provide a camera module and a terminal device. The camera module is an ultra wide-angle lens, and a non-rotationally symmetric free-form lens is introduced into the lens, to resolve an optical distortion problem of the ultra wide-angle lens, implement an ultra wide-angle and low-distortion imaging effect, and provide a user with good experience.

(7) According to a first aspect, this embodiment provides a camera module, applied to a terminal device, and includes a plurality of lenses sequentially arranged from an object side to an image side along a direction of an optical axis. A quantity of the lenses may be three, four, five, six, seven, or the like. At least one lens of the plurality of lenses is a free-form lens. The free-form lens is a non-rotationally symmetric lens, the 1st lens in the plurality of lenses in a direction from the object side to the image side is a first lens, a distance on the optical axis between an object-side surface of the first lens and an imaging surface is TTL, an effective focal length of the camera module is EFL, and $TTL/EFL \leq 2.0$, to implement relatively short TTL. The distance on the optical axis between the object side surface of the first lens and the imaging surface is TTL, which is specifically a distance between an intersection of an object-side surface of the first lens and the optical axis and an intersection of the imaging surface and the optical axis. In this application a surface type of the free-form lens is non-rotationally symmetric. The non-rotationally symmetric free-form lens can increase freedom of optical design of the camera module, and a rectangular imaging region can be implemented without limiting an imaging region to being a rotationally symmetric image circle. An electronic image sensor is arranged on the imaging surface, and is specifically a sensor chip of a camera in the terminal device. In this application, at least one lens is limited to being a free-form lens, to suppress an optical distortion problem of the camera module. An imaging effect of the camera module can be ensured even in the case of an ultra wide angle. In addition, a relatively short total length of the camera module can be obtained. In other words, the non-rotationally symmetric free-form lens is introduced into the camera module, so that an ultra wide-angle and low-distortion imaging effect can be implemented, that is, a system aberration of the camera module can be reduced or minimized, to implement functions of aberration correction and distortion reduction. In addition, the free-form lens can further reduce TTL of the camera module, so that the camera module has a compact structure.

(8) In a possible implementation, an X-axis and the optical axis form a first plane, a Y-axis and the optical axis form a second plane, and the X-axis and the Y-axis are two central axes that are perpendicular to each other on the imaging surface of the camera module. An intersection of the X-axis and the Y-axis is located on the optical axis. The free-form lens is a centrosymmetric structure by using the first plane as a center, and the free-form lens is also a centrosymmetric structure by using the second plane as a center. Symmetry of the free-form lens in a direction of the X-axis and in a direction of the Y-axis helps ensure imaging quality. The imaging region of the electronic sensor is rectangular. Symmetry of the free-form lens in the direction of the X-axis and in the direction of the Y-axis makes imaging quality of the imaging region keep particular symmetry. This helps implement that imaging quality of a middle region close to the optical axis is better than imaging quality of an edge region away from the optical axis.

(9) In a possible implementation, an object-side surface or an image-side surface of the free-form lens or both are free-form surfaces, and a surface type expression of the free-form surface is:

$$(10) z = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + \sum_{i=1}^M \text{Math}. A_i E_i$$

(11) z is a sag of an optical surface, and z is an expression of x and y; k is a conic coefficient; c is a curvature radius; r is a radius height in the direction of the optical axis; $r_{sup.2} = x_{sup.2} + y_{sup.2}$; A_{sub.i} is a polynomial coefficient; and E_{sub.i} is a monomial of an X-axis coordinate and a Y-axis coordinate. Directions of the x-axis and the y-axis herein are consistent with the foregoing directions of the X-axis and the Y-axis about which the free-form lens is symmetric.

$$(12) \sum_{i=1}^M \text{Math}. A_i E_i = A_1 x^0 y^2 + A_2 x^2 y^0 + A_3 x^2 y^2 + A_4 x^4 y^0 + A_5 x^0 y^4 + A_6 x^6 y^0 + A_7 x^0 y^6 + A_8 x^4 y^2 + A_9 x^4 y^2 + \text{Math}.$$

(13) Exponents of x and y in E_{sub.i} of the surface type expression of the free-form lens are both even numbers, x is an X-axis coordinate, and y is a Y-axis coordinate, so that the surface type of the free-form lens has symmetry. Specifically, the exponents of x and y in E_{sub.i} of the surface type expression of free-form lens are both even numbers. This can make the surface type of the free-form lens better symmetric, and facilitate lens processing and detection.

(14) In a second implementation, an object-side surface or an image-side surface of the free-form lens or both are free-form surfaces, and a surface type expression of the free-form lens is:

$$(15) z = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + \sum_{i=1}^M \text{Math}. A_i E_i$$

(16) z is a sag of an optical surface, and z is an expression of x and y; k is a conic coefficient; c is a curvature radius; r is a radius height in the direction of the optical axis; $r_{sup.2} = x_{sup.2} + y_{sup.2}$; A_{sub.i} is a polynomial coefficient; and E_{sub.i} is a monomial of an X-axis coordinate and a Y-axis coordinate.

$$(17) \sum_{i=1}^M \text{Math}. A_i E_i = A_1 \text{Math}. x^1 y^0 \text{Math}. + A_2 \text{Math}. x^0 y^1 \text{Math}. + A_3 \text{Math}. x^1 y^1 \text{Math}. + A_4 \text{Math}. x^0 y^2 \text{Math}. + A_5 \text{Math}. x^3 y^0 \text{Math}. + A_6 \text{Math}.$$

(18) A_{sub.i} is a polynomial coefficient, x is an X-axis coordinate, and y is a Y-axis coordinate.

(19) In a third implementation, an object-side surface or an image-side surface of the free-form lens or both are free-form surfaces, and a surface type expression of the free-form lens is:

$$(20) z = \frac{c_x x^2 + c_y y^2}{1 + \sqrt{1 - (1+k_x)c_x^2 x^2 - (1+k_y)c_y^2 y^2}} + \sum_{i=1}^M \text{Math}. A_i \text{Math}. x^i \text{Math}. + \sum_{i=1}^M \text{Math}. B_i \text{Math}. y^i \text{Math}.$$

(21) z is a sag of an optical surface, and z is an expression of x and y; x is an x-axis coordinate, and y is a Y-axis coordinate; k_{sub.x} and k_{sub.y} are conic coefficients; c_{sub.x} and c_{sub.y} are curvature radii; and A_{sub.i} and B_{sub.i} are polynomial coefficients.

(22) In a possible implementation, the X-axis and the Y-axis are two central axes passing through a center of the imaging surface and respectively parallel to a long side and a short side of the imaging surface.

(23) In a possible implementation, a quantity of the plurality of lenses is N, N ≥ 3, and the plurality of lenses include the first lens to the Nth lens sequentially arranged in the direction from the object side to the image side; and surface types of object-side surfaces and image-side surfaces of the first lens to the (N—1)th lens are all aspheric surfaces, and the Nth lens is a free-form lens. A quantity of lenses used in an embodiment of this application is six. In another implementation, the quantity of lenses may be seven, eight, or the like. In an actual implementation process, a camera module including three, four, or five lenses may be disposed as required. The free-form lens is introduced into the camera module, so that a visual effect without image distortion is achieved for an ultra wide-angle lens set, and imaging quality unachievable by ordinary aspheric optical design can be achieved.

(24) In a possible implementation, a half of a diagonal length of an effective pixel region of the imaging surface of the camera module is ImgH, and $TTL/ImgH \leq 2.0$, to help limit a total system length of the camera module, so that the camera module has a compact structure and meets a design

requirement for a portable device.

(25) In a possible implementation, an entrance pupil diameter of the camera module is EPD, and $EFL/EPD \leq 2.2$. In this implementation, $EPD/EPD \leq 2.2$ is used for limitation, so that an f-number of the camera module is limited, which is conducive to imaging quality.

(26) In a possible implementation, a field of view of the camera module is FOV, $FOV \geq 100$ deg, and $EFL < 20$ mm. The camera module has an ultra wide-angle imaging effect. In this implementation, the free-form lens is introduced into the camera module, to achieve a short-focal-length, large-field-of-view, and low-distortion imaging effect for the ultra wide-angle lens set. In addition, a relatively short total length of the camera module is obtained, and a compact structure of the ultra wide-angle lens set is implemented.

(27) In a possible implementation, the quantity of the plurality of lenses is N, and the first three lenses arranged sequentially in the direction from the object side to the image side are respectively the first lens, a second lens, and a third lens. The camera module further includes a vignetting stop, and the vignetting stop is disposed on an object side of the second lens or on an object side of the third lens. The vignetting stop can intercept an imaging ray with a large aberration (also referred to as intercepting a ray of a point beyond the axis), to improve imaging quality. The vignetting stop is arranged close to a middle position of the camera module, to help balance a comprehensive aberration of the camera module.

(28) In a possible implementation, the lens adjacent to the imaging surface in the plurality of lenses is the free-form lens, a curvature radius of the object-side surface of the free-form lens is R61, and a curvature radius of the image-side surface of the free-form lens is R62. The following condition is met: $|f/R61| + |f/R62| < 2$, to help correct the comprehensive aberration of the camera set. When $|f/R61| + |f/R62|$ is greater than or equal to 2, it does not help correct the comprehensive aberration of the camera module, and an imaging effect is affected.

(29) In a possible implementation, the camera module further includes an infrared filter element located between the plurality of lenses and the imaging surface. The infrared filter element can effectively alleviate color cast generated around an image.

(30) Optical design of the free-form lens can reduce the quantity of lenses of the camera module, and decrease a weight of the camera module, thereby facilitating a lightweight implementation. A flexible spatial layout and design freedom of the free-form lens simplify a structure of the camera module. The optical design of the free-form lens increases freedom of optimization, and helps control the system aberration of the camera module and improve overall quality of the camera module.

(31) In a possible implementation, the camera module further includes an electronic image sensor, the electronic image sensor is disposed on the imaging surface, and the imaging surface of the camera module is a rectangular region, which matches an image sensing area of the electronic image sensor and is not less than the image sensing area of the electronic image sensor. In other words, the electronic image sensor is rectangular, and the imaging surface of the camera module using the free-form lens is a rectangular region, to match the image sensor, and improve imaging quality.

(32) In a possible implementation, a diagonal length of the image sensing surface of the electronic image sensor is not less than 5.5 mm. For image sensors with same effective pixels, generally, a larger size of an image sensor indicates a larger unit area of each pixel, higher performance, more image details that can be recorded. In this implementation, the diagonal length of the image sensing surface is limited to being not less than 5.5 mm. This can ensure that the camera module applied to a mobile terminal has high image sensing performance, and provides high image quality.

(33) According to a second aspect, this application provides a terminal device, including the camera module according to any one of the foregoing implementations.

(34) In this application, the non-rotationally symmetric free-form lens is introduced into the lens set, to resolve an optical distortion problem of the ultra wide-angle lens set, eliminate impact of imaging distortion of the camera module, improve user experience, and obtain excellent optical quality.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) To describe the technical solutions in the embodiments of this application or the background more clearly, the following describes the accompanying drawings required for use in the embodiments of this application or the background.

(2) FIG. 1 is a schematic diagram of a camera module applied to a terminal device according to this application;

(3) FIG. 1a is a schematic diagram of a free-form lens that has line symmetry in a direction of an X-axis and in a direction of a Y-axis;

(4) FIG. 2a and FIG. 2b are schematic diagrams of a camera module according to Embodiment 1 of this application;

(5) FIG. 2c is a distortion curve of an optical system in Embodiment 1;

(6) FIG. 2d is a lateral chromatic aberration curve of an optical system in Embodiment 1;

(7) FIG. 3a and FIG. 3b are schematic diagrams of a camera module according to Embodiment 2 of this application;

(8) FIG. 3c is a distortion curve of an optical system in Embodiment 2;

(9) FIG. 3d is a lateral chromatic aberration curve of an optical system in Embodiment 2;

(10) FIG. 4a and FIG. 4b are schematic diagrams of a camera module according to Embodiment 3 of this application;

(11) FIG. 4c is a distortion curve of an optical system in Embodiment 3; and

(12) FIG. 4d is a lateral chromatic aberration curve of an optical system in Embodiment 3.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(13) The following describes the embodiments of this application with reference to the accompanying drawings in the embodiments of this application.

(14) Refer to FIG. 1. A camera module 10 in this application is applied to a terminal device 100. The terminal device 100 may be a portable terminal such as a mobile phone or a tablet, and the camera module 10 may be an ultra wide-angle lens set. The camera module 10 is assembled inside the terminal device 100, and may be a rear camera or a front camera of the terminal device 100, or a retractable camera that may extend out of a housing of the terminal device 100.

(15) In an implementation, the camera module provided in this application includes six lenses (six lenses are used as a specific embodiment for description, and a quantity of lenses is not limited in this application). The six lenses are sequentially distributed from an object side to an image side along a direction of an optical axis as follows: a first lens, a second lens, a third lens, a fourth lens, a fifth lens, and a sixth lens. The quantity of the lenses is not limited to six, and may be another quantity, such as three, four, five, seven. At least one lens of the plurality of lenses is a free-form lens, and the free-form lens is a non-rotationally symmetric surface type. The non-rotationally symmetric free-form lens can increase freedom of optical design of the camera module. A distance on the optical axis between an object-side surface of the first lens and an imaging surface is TTL, an effective focal length of the camera module is EFL, and $TTL/EFL \leq 2.0$, to implement relatively short TTL, thereby facilitating a compact structure of the camera module. At least one lens is limited to being a non-rotationally symmetric free-form surface, so that an optical distortion problem of the ultra wide-angle lens set can be alleviated, and the camera module has a compact structure, thereby improving user experience. Aspheric curve equation of lenses 1 to 5 is as follows:

$$(16) z = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + \sum_{i=1}^M \text{Math}_i \cdot r^i$$

(17) z is a sag of an optical surface, and z is an expression of x and y; k is a conic coefficient; c is a curvature radius; r is a radius height in the direction of the optical axis; $r.\text{sup}.2 = x.\text{sup}.2 + y.\text{sup}.2$; x is an X-axis coordinate, and y is a Y-axis coordinate; Math_i is a polynomial coefficient; and $p.\text{sub}.i$ is a normalized radial coordinate.

(18) In this application, a surface type expression (three different surface type expressions are listed below) of the free-form lens is defined, to implement symmetry of the free-form lens in a direction of an X-axis and a direction of a Y-axis. The direction of the X-axis and the direction of the

Y-axis are two directions perpendicular to each other on the imaging surface of the camera module.

(19) In a first implementation, a surface type expression (namely, a free-form sphere curve equation) of the non-rotationally symmetric free-form lens is expressed as follows:

$$(20) z = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + \sum_{i=1}^M \text{Math}. A_i E_i$$

(21) z is a sag of an optical surface, and z is an expression of x and y; k is a conic coefficient; c is a curvature radius; r is a radius height in the direction of the optical axis; $r.\text{sup}.2 = x.\text{sup}.2 + y.\text{sup}.2$; $A.\text{sub}.i$ is a polynomial coefficient; and $E.\text{sub}.i$ is a monomial of an X-axis coordinate and a Y-axis coordinate. Directions of the x-axis and the y-axis herein are consistent with the foregoing directions of the X-axis and the Y-axis about which the free-form lens is symmetric.

$$(22) \sum_{i=1}^N \text{Math}. A_i E_i = A_1 x^0 y^2 + A_2 x^2 y^0 + A_3 x^2 y^2 + A_4 x^4 y^0 + A_5 x^0 y^4 + A_6 x^6 y^0 + A_7 x^0 y^6 + A_8 x^4 y^2 + A_9 x^4 y^2 + \text{Math}.$$

(23) $A.\text{sub}.i$ is a polynomial coefficient.

(24) Exponents of x and y in $E.\text{sub}.i$ of the surface type expression of the free-form lens are both even numbers, x is an X-axis coordinate, and y is a Y-axis coordinate, so that the surface type of the free-form lens has symmetry.

(25) In a second implementation, a surface type expression of the free-form lens is:

$$(26) z = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + \sum_{i=1}^M \text{Math}. A_i E_i$$

(27) z is a sag of an optical surface, and z is an expression of x and y; k is a conic coefficient; c is a curvature radius; r is a radius height in the direction of the optical axis; $r.\text{sup}.2 = x.\text{sup}.2 + y.\text{sup}.2$; $A.\text{sub}.i$ is a polynomial coefficient; and $E.\text{sub}.i$ is a monomial of an X-axis coordinate and a Y-axis coordinate.

(28) 0

$$\sum_{i=1}^N \text{Math}. A_i E_i = A_1 \text{Math}. x^1 y^0 \text{Math}. + A_2 \text{Math}. x^0 y^1 \text{Math}. + A_3 \text{Math}. x^1 y^1 \text{Math}. + A_4 \text{Math}. x^0 y^2 \text{Math}. + A_5 \text{Math}. x^3 y^0 \text{Math}. + A_6 \text{Math}.$$

(29) $A.\text{sub}.i$ is a polynomial coefficient, x is an X-axis coordinate, and y is a Y-axis coordinate.

(30) In a third implementation, a surface type expression of the free-form lens is:

$$(31) z = \frac{c_x x^2 + c_y y^2}{1 + \sqrt{1 - (1+k_x)c_x^2 x^2 - (1+k_y)c_y^2 y^2}} + \sum_{i=1}^M \text{Math}. A_i \text{Math}. x^i \text{Math}. + \sum_{i=1}^M \text{Math}. B_i \text{Math}. y^i \text{Math}.$$

(32) z is a sag of an optical surface; x is an X-axis coordinate, and y is a Y-axis coordinate; $k.\text{sub}.x$ and $k.\text{sub}.y$ are conic coefficients; $c.\text{sub}.x$ and $c.\text{sub}.y$ are curvature radii; and $A.\text{sub}.i$ and $B.\text{sub}.i$ are polynomial coefficients.

(33) In the foregoing implementations, the free-form lens may have one surface that is a free-form surface, for example, an object-side surface or an image-side surface is a free-form surface, or both the surfaces may be free-form surfaces, that is, both the object-side surface and the image-side surface are free-form surfaces.

(34) In an implementation, FIG. 1a is a schematic diagram of a free-form lens that is symmetric in a direction of an X-axis and in a direction of a Y-axis, an imaging surface S14 is a rectangular region, and the X-axis and the Y-axis are two central axes perpendicular to each other on the imaging surface S14. An intersection of the X-axis and the Y-axis is located on an optical axis. Specifically, the direction of the X-axis is a central axis that passes through a center of the rectangular imaging surface S14 and is parallel to a long side of the rectangular imaging surface S14. The direction of the Y-axis is a central axis that passes through the center of the rectangular imaging surface S14 and is parallel to a short side of the rectangular imaging surface S14. The X-axis and the optical axis form a first plane, and the Y-axis and the optical axis form a second plane. The free-form lens L6 is a centrosymmetric structure by using the first plane as a center, and the free-form lens is also a centrosymmetric structure by using the second plane as a center. Keeping the free-form lens L6 centrosymmetric with respect to the first plane and centrosymmetric with respect to the second plane helps ensure imaging quality, and helps implement that imaging quality of a middle region close to the optical axis is better than imaging quality of an edge region away from the optical axis.

(35) The following describes this application in detail by using three specific embodiments.

Embodiment 1

(36) As shown in FIG. 2a and FIG. 2b, a straight line in the middle represents an optical axis, a left side of a camera module is an object side, and a right side of the camera module is an image side. In the camera module provided in this embodiment, a first lens L1, a stop STO, a second lens L2, a third lens L3, a fourth lens L4, a fifth lens L5, a sixth lens L6, an infrared filter element IRCF, and an electronic image sensor are sequentially arranged along the optical axis from the object side to the image side. The electronic image sensor may be placed at a position of an imaging surface S14. In this implementation, the stop STO is placed after the first lens L1, and is close to a middle position of the camera module, to help balance an aberration of the camera module.

(37) The first lens L1 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S1 of the first lens L1 is convex. A region, near the optical axis, of an image-side surface S2 of the first lens L1 is convex. Both the regions are aspheric surfaces.

(38) The second lens L2 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S3 of the second lens L2 is convex. A region, near the optical axis, of an image-side surface S4 of the second lens L2 is concave. Both the regions are aspheric surfaces.

(39) The third lens L3 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S5 of the third lens L3 is concave. A region, near the optical axis, of an image-side surface S6 of the third lens L3 is concave. Both the regions are aspheric surfaces.

(40) The fourth lens L4 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S7 of the fourth lens L4 is concave. A region, near the optical axis, of an image-side surface S8 of the fourth lens L4 is concave. Both the regions are aspheric surfaces.

(41) The fifth lens L5 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S9 of the fifth lens L5 is concave. A region, near the optical axis, of an image-side surface S10 of the fifth lens L5 is concave. Both the regions are aspheric surfaces.

(42) The sixth lens L6 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S11 of the sixth lens L6 is concave. A region, near the optical axis, of an image-side surface S12 of the sixth lens L6 is convex. Both the regions are free-form surfaces.

(43) An object-side surface S13 and an image-side surface S14 of the infrared filter element IRCF are both flat surfaces.

(44) In Embodiment 1, a distance on the optical axis between the object-side surface S1 of the first lens L1 and an imaging surface S14 of an infinitely far-away object is TTL, and an effective focal length of the camera module is EFL. The following condition can be met: $TTL/EFL \leq 2.0$, to implement shorter TTL, thereby facilitating miniaturization design of the camera module, saving internal space of a terminal device, and facilitating thinning development of the terminal device.

(45) In Embodiment 1, the exponents of x and y in $E.\text{sub}.i$ of the surface type expression of the free-form surface of the sixth lens L6 are both even numbers. This can make the surface type of the lens better symmetric, and facilitate lens processing and detection.

(46) In an extension of Embodiment 1, optionally, a vignetting stop ST1 (not shown) may be disposed before (namely, on the object side of) the first

lens L1, and a vignetting stop ST2 (not shown) may be disposed after (on the image side of) the sixth lens L6, to effectively reduce a diameter of the camera module.

(47) In Embodiment 1, a focal length of the camera module is f , a curvature radius of the object-side surface S11 of the sixth lens L6 is R61, and a curvature radius of the image-side surface S12 of the sixth lens L6 is R62. The following condition is met: $|f/R61|+|f/R62|=1.37$, to help correct a comprehensive aberration of a camera set, so that a lateral chromatic aberration of the camera module is less than 2%.

(48) Table 1a is a table showing characteristics of an optical system in this embodiment. A curvature radius and a thickness are both expressed in millimeters (mm).

(49) TABLE-US-00001

TABLE 1a	Surface	Surface Curvature	Refractive	Dispersion	number type	radius	Thickness	Material	index	coefficient
S1	Aspheric	5.5545	0.2145	Resin	1.65	22.9	surface	S2	Aspheric	43.3006
								STO	Flat	Infinite
								S3	Aspheric	72.4020
								S4	Aspheric	-2.2277
								S5	Aspheric	-4.1395
								S6	Aspheric	-6.0243
								S7	Aspheric	-12.8096
								S8	Aspheric	-1.5405
								S9	Aspheric	-1.3990
								S10	Aspheric	-1.7201
								S11	Extended	-3.2543
								S12	Extended	0.5324
								S13	Flat	Infinite
								S14	Flat	Infinite

(50) Table 1b gives conic coefficients k and polynomial coefficients $a1, a2, a3, a4, a5, a6, a7, a8, a9, a10, a11, a12, a13, a14$, and $a15$ that can be used for aspheric lenses surfaces S1 to S10 in Embodiment 1.

(51) TABLE-US-00002

TABLE 1b	Parameter	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	K	-89.0714	130.6859	-4655.9692	-0.5553	6.5588	0.9185	45.9472	-0.7741	-0.6498	-0.4395	a1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	a2	-0.0152	0.2774	0.1147	-0.0006	-0.1417	0.1306	-0.1796	-0.7795	-0.8846	-0.6002	a3	0.3160	0.3257	-0.0143	0.0382	-0.0718	-0.1298	-0.0613	0.2977	0.7433	0.6135	a4	0.1162	-0.2224	-0.1036	-0.7000	-0.1193	0.0432	0.1270	0.4062	0.6841	0.3106	a5	1.3304	1.3382	-0.0501	0.5297	-0.1721	0.1261	-0.3324	0.2381	0.0375	-0.3944	a6	-0.2317	3.3063	-0.0793	-0.6215	-0.1170	0.0307	-0.2253	-0.0309	-0.0484	-0.0173	a7	-1.3109	-0.8758	-0.4983	-0.1927	-0.0370	-0.0910	0.2503	-0.2238	-0.0885	0.0731	a8	0.6204	-11.8832	-0.4727	-0.3596	-0.0438	-0.1312	0.7093	-0.2444	-0.2185	0.1842	a9	-0.8465	-2.1909	0.3526	-1.7055	-0.1584	-0.1027	0.2254	-0.1653	0.0489	0.0459	a10	-0.6141	41.7733	-4.2841	-0.1863	-0.2957	-0.0327	-0.6322	-0.0264	0.1733	-0.0481	a11	3.1749	-7.4454	1.0910	-1.6257	-0.3721	0.0085	-0.1870	0.0887	0.0527	0.0010	a12	5.3311	-45.5026	4.8330	-4.4567	-0.2846	0.0541	-0.1634	0.1337	0.0051	-0.0070	a13	3.6623	33.6188	-0.5357	15.8836	0.0727	0.0683	0.0005	0.1275	-0.0144	-0.0087	a14	-0.9079	85.3587	15.6750	19.4730	0.7811	0.0624	0.2713	0.0584	-0.0541	-0.0043	a15	-12.2109	97.2852	-157.7664	-8.8749	1.9354	0.0098	0.4500	-0.0086	-0.0277	0.0034
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(52) Table 1c gives conic coefficients k and higher-order term coefficients X2Y0, X0Y2, X4Y0, X2Y2, X0Y4, X6Y0, X4Y2, X2Y4, X0Y6, X8Y0, X6Y2, X4Y4, X2Y6, and X0Y8 that can be used for the free-form surfaces S11 and S12 in Embodiment 1.

(53) TABLE-US-00003

TABLE 1c	Parameter	S11	S12	K	-0.2054	-1.4469	X2Y0	1.7232	-1.0737	X0Y2	1.7480	-1.0137	X4Y0	-1.5261	0.0639	X2Y2	-3.0799	-0.0764	X0Y4	-1.4951	-0.0559	X6Y0	1.1911	-0.0397	X4Y2	3.4068	0.2676	X2Y4	3.0357	0.3088	X0Y6	0.8664	0.0968	X8Y0	-0.7572	0.0372	X6Y2	-2.6574	-0.1552	X4Y4	-2.2974	-0.2318	X2Y6	-1.3632	-0.2295	X0Y8	-0.1705	-0.0414
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(54) FIG. 2C shows a distortion curve of the optical system in Embodiment 1, which represents distortion values corresponding to different fields of view.

(55) FIG. 2d shows a lateral chromatic aberration curve of the optical system in Embodiment 1, which represents lateral chromatic aberration values corresponding to five different wavelengths of light at different fields of view. Arrow indication lines are used to represent the five different wavelengths of light. The wavelengths are 510 nanometers, 470 nanometers, 610 nanometers, 550 nanometers, and 650 nanometers, respectively.

(56) It can be learned from FIG. 2c and FIG. 2d that the optical system provided in Embodiment 1 can achieve good imaging quality.

Embodiment 2

(57) As shown in FIG. 3a and FIG. 3b, in a camera module in this implementation, a first lens L1, a second lens L2, a stop STO, a third lens L3, a fourth lens L4, and a fifth lens L5, a sixth lens L6, an infrared filter element IRCF, and an electronic image sensor are sequentially arranged along an optical axis from an object side to an image side. The electronic image sensor can be placed on a position of an imaging surface S14 (also referred to as an image surface). In this implementation, the stop STO is placed after the second lens L2, and is close to a middle position of the camera module, to help balance an aberration of the camera module.

(58) The first lens L1 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S1 of the first lens L1 is concave. A region, near the optical axis, of an image-side surface S2 of the first lens L1 is convex. Both the regions are aspheric surfaces.

(59) The second lens L2 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S3 of the second lens L2 is convex. A region, near the optical axis, of an image-side surface S4 of the second lens L2 is convex. Both the regions are aspheric surfaces.

(60) The third lens L3 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S5 of the third lens L3 is convex. A region, near the optical axis, of an image-side surface S6 of the third lens L3 is concave. Both the regions are aspheric surfaces.

(61) The fourth lens L4 has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S7 of the fourth lens L4 is concave. A region, near the optical axis, of an image-side surface S8 of the fourth lens L4 is concave. Both the regions are aspheric surfaces.

(62) The fifth lens L5 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S9 of the fifth lens L5 is concave. A region, near the optical axis, of an image-side surface S10 of the fifth lens L5 is concave. Both the regions are aspheric surfaces.

(63) The sixth lens L6 has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface S11 of the sixth lens L6 is concave. A region, near the optical axis, of an image-side surface S12 of the sixth lens L6 is convex. Both the regions are free-form surfaces.

(64) In Embodiment 2, a distance on the optical axis between the object-side surface S1 of the first lens L1 and an imaging surface S14 of an infinitely far-away object is TTL, and an effective focal length of the camera module is EFL. The following condition can be met: $TTL/EFL \leq 2$, to implement shorter TTL, thereby facilitating miniaturization design of the camera module, saving internal space of a terminal device, and facilitating thinning development of the terminal device.

(65) In Embodiment 2, the exponents of x and y in E.sub.i of the surface type expression of the free-form surface of the sixth lens L6 are both even numbers. This can make the surface type of the lens better symmetric, and facilitate lens processing and detection.

(66) In an extension of Embodiment 2, optionally, a vignetting stop ST1 (not shown) may be disposed before (namely, on the object side of) the first lens L1, and a vignetting stop ST2 (not shown) may be disposed after (on the image side of) the sixth lens L6, to effectively reduce a diameter of the camera module.

(67) In Embodiment 2, a focal length of the camera module is f , a curvature radius of the object-side surface S11 of the sixth lens L6 is R61, and a curvature radius of the image-side surface S12 of the sixth lens L6 is R62. The following condition is met: $|f/R61|+|f/R62|=0.71$, to help correct a comprehensive aberration of a camera set, so that a lateral chromatic aberration of the camera module is less than 3 μm and a distortion is less than 2%.

(68) Table 2a is a table showing characteristics of an optical system in this embodiment. A curvature radius and a thickness are both expressed in

millimeters (mm).

(69) TABLE-US-00004 TABLE 2a Surface Surface Curvature Refractive Dispersion number type radius Thickness Material index coefficient
Material S1 Aspheric -2.3847 0.3458 Resin 1.65 22.9 EP7000 surface S2 Aspheric 1.8127 0.0363 surface S3 Aspheric 1.458 0.2855 Resin 1.67
19.243 EP9000 surface S4 Aspheric 3.0075 0.5178 surface STO Flat Infinite -0.0091 surface S5 Aspheric 47.486 0.7245 Resin 1.54 55.99
APL5014CL surface S6 Aspheric -2.3808 0.6004 surface S7 Aspheric -7.4388 0.9962 Resin 1.54 55.99 APL5014CL surface S8 Aspheric -1.4872
0.096 surface S9 Aspheric -1.0653 0.2928 Resin 1.67 19.243 EP9000 surface S10 Aspheric -1.6155 0.0363 surface S11 Extended -0.2474 0.9056
Resin 1.65 22.9 EP7000 aspheric surface S12 Extended 1.6942 1.2628 aspheric surface S13 Flat Infinite 0.2096 Glass 1.52 54.5 D263T surface Flat
Infinite 0.0499 surface S14 Flat Infinite 0 surface

(70) Table 2b gives conic coefficients k and polynomial coefficients $a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9, a_{10}, a_{11}, a_{12}, a_{13}, a_{14}$, and a_{15} that can be used for aspheric lenses surfaces **S1** to **S10** in Embodiment 2.

(71) TABLE-US-00005 TABLE 2b Parameter S1 S2 S3 S4 S5 S6 S7 S8 S9 S10 K -20.2834 -2.3818 -6.3801 -57.3023 -100.0000 0.3347 23.0097
-0.5407 -0.6057 -0.7241 a_1 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 a_2 0.3807 -1.1425 -0.0770 0.6250 0.1903
0.1292 -0.4347 -0.8528 -0.9073 -0.5643 a_3 0.0540 1.0089 0.8594 0.2524 0.0154 0.0458 0.2969 1.1047 1.4486 0.3965 a_4 0.2938 -0.0461 -0.9054
-0.1618 -0.1345 -0.4954 -0.5458 -0.2717 0.7216 0.8305 a_5 -0.0688 -0.0713 0.6027 -0.0884 0.0204 0.6727 -0.1883 0.1518 -0.3589 -0.5759 a_6
-0.0549 -0.3815 -0.8218 1.7311 0.0268 -0.9452 0.2630 0.0884 -0.0851 -0.2502 a_7 -0.0334 0.3001 -1.0009 -3.0152 -0.3675 -0.5039 0.3214
-0.1407 -0.1323 0.0253 a_8 0.0225 0.3473 1.0333 -8.5801 -0.5188 1.2010 0.5801 -0.3391 -0.1651 0.2075 a_9 0.0821 -0.3709 -0.5461 -0.0307
0.4198 0.1419 0.0467 -0.2976 -0.0121 0.1278 a_{10} 0.0606 0.1619 -0.6811 47.1165 -3.5227 0.0874 -0.7888 -0.1074 -0.0245 0.0155 a_{11} -0.0499
-0.1433 1.7402 -1.9290 3.3752 -5.1732 -0.3589 -0.0821 -0.0951 0.0291 a_{12} -0.0466 -0.0746 1.4197 -54.5390 10.0663 -10.3682 -0.1919
-0.0458 -0.0297 -0.0061 a_{13} -0.0083 0.0858 1.4019 -16.3711 10.5950 13.9387 -0.3328 -0.1357 -0.0921 -0.0237 a_{14} 0.0474 -0.2141 -0.1854
-50.7892 -12.5135 25.7124 0.5221 -0.0465 0.3056 -0.0251 a_{15} 0.0322 -0.2401 -1.3685 15.2032 -156.0708 9.4206 0.7632 0.2883 0.4513
-0.0132 a_{16} -0.0819 0.4074 -2.0867 193.8807 170.3787 -57.1594 1.0040 0.9630 0.3100 0.0046

(72) Table 2C gives conic coefficients k and higher-order term coefficients $X_2Y_0, X_0Y_2, X_4Y_0, X_2Y_2, X_0Y_4, X_6Y_0, X_4Y_2, X_2Y_4, X_0Y_6, X_8Y_0, X_6Y_2, X_4Y_4, X_2Y_6$, and X_0Y_8 that can be used for the free-form surfaces **S11** and **S12** in Embodiment 2.

(73) TABLE-US-00006 TABLE 2c Parameter S11 S12 K -1.0178 -4.3499 X_2Y_0 6.5795 0.0633 X_0Y_2 6.5725 0.0489 X_4Y_0 -1.725 -0.2768 X_2Y_2
-3.4771 -0.5611 X_0Y_4 -1.7136 -0.2535 X_6Y_0 0.4593 0.1003 X_4Y_2 1.6453 0.4314 X_2Y_4 1.527 0.3351 X_0Y_6 0.4794 0.0951 X_8Y_0 0.1257 0.0157
 X_6Y_2 -0.2977 -0.2672 X_4Y_4 0.028 -0.1712 X_2Y_6 -0.0265 -0.1196 X_0Y_8 0.0337 -0.0193

(74) FIG. 3c shows a distortion curve of the optical system in Embodiment 2, which represents distortion values corresponding to different fields of view.

(75) FIG. 3d shows a lateral chromatic aberration curve of the optical system in Embodiment 2, which represents lateral chromatic aberration values corresponding to five different wavelengths of light at different fields of view. Arrow indication lines are used to represent the five different wavelengths of light. The wavelengths are 510 nanometers, 470 nanometers, 610 nanometers, 550 nanometers, and 650 nanometers, respectively.

(76) It can be learned from FIG. 3c and FIG. 3d that the optical system provided in Embodiment 2 can achieve good imaging quality.

Embodiment 3

(77) As shown in FIG. 4a and FIG. 4b, in a camera module in this implementation, a first lens **L1**, a second lens **L2**, a stop **STO**, a third lens **L3**, a fourth lens **L4**, and a fifth lens **L5**, a sixth lens **L6**, an infrared filter element **IRCF**, and an electronic image sensor are sequentially arranged along an optical axis from an object side to an image side. The electronic image sensor can be placed on a position of an imaging surface **S14**.

(78) The first lens **L1** has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface **S1** of the first lens **L1** is concave. A region, near the optical axis, of an image-side surface **S2** of the first lens **L1** is convex. Both the regions are aspheric surfaces.

(79) The second lens **L2** has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface **S3** of the second lens **L2** is convex. A region, near the optical axis, of an image-side surface **S4** of the second lens **L2** is convex. Both the regions are aspheric surfaces.

(80) The third lens **L3** has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface **S5** of the third lens **L3** is convex. A region, near the optical axis, of an image-side surface **S6** of the third lens **L3** is concave. Both the regions are aspheric surfaces.

(81) The fourth lens **L4** has a positive refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface **S7** of the fourth lens **L4** is concave. A region, near the optical axis, of an image-side surface **S8** of the fourth lens **L4** is concave. Both the regions are aspheric surfaces.

(82) The fifth lens **L5** has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface **S9** of the fifth lens **L5** is concave. A region, near the optical axis, of an image-side surface **S10** of the fifth lens **L5** is concave. Both the regions are aspheric surfaces.

(83) The sixth lens **L6** has a negative refractive power and is made of a resin material. A region, near the optical axis, of an object-side surface **S11** of the sixth lens **L6** is concave. A region, near the optical axis, of an image-side surface **S12** of the sixth lens **L6** is convex. Both the regions are free-form surfaces.

(84) In Embodiment 3, a distance on the optical axis between the object-side surface **S1** of the first lens **L1** and an imaging surface **S14** of an infinitely far-away object is TTL, and an effective focal length of the camera module is EFL. The following condition can be met: $TTL/EFL \leq 2$, to implement shorter TTL, thereby facilitating miniaturization design of the camera module, saving internal space of a terminal device, and facilitating thinning development of the terminal device.

(85) In Embodiment 3, the exponents of x and y in $E_{sub.i}$ of the surface type expression of the free-form surface of the sixth lens **L6** are both even numbers, which makes the surface type of the lens better symmetric, and facilitates lens processing and detection.

(86) In an extension of Embodiment 3, optionally, a vignetting stop **ST1** (not shown) may be disposed before (namely, on the object side of) the first lens **L1**, and a vignetting stop **ST2** (not shown) may be disposed after (on the image side of) the sixth lens **L6**, to effectively reduce a diameter of the camera module.

(87) In Embodiment 3, a focal length of the camera module is f , a curvature radius of the object-side surface **S11** of the sixth lens **L6** is R_{61} , and a curvature radius of the image-side surface **S12** of the sixth lens **L6** is R_{62} . The following condition is met: $|f/R_{61}| + |f/R_{62}| = 1.92$, to help correct a comprehensive aberration of a camera set, so that a lateral chromatic aberration of the camera module is less than 3 μm and a distortion is less than 2%.

(88) Table 3a is a table showing characteristics of an optical system in this embodiment. A curvature radius and a thickness are both expressed in millimeters (mm).

(89) TABLE-US-00007 TABLE 3a Surface Surface Curvature Refractive Dispersion number type radius Thickness Material index coefficient
Material S1 Aspheric -3.1589 0.2953 Resin 1.65 22.9 EP7000 surface S2 Aspheric 3.5364 0.0237 surface S3 Aspheric 1.7330 0.2483 Resin 1.65
22.9 EP7000 surface S4 Aspheric 3.3108 0.4889 surface STO Flat Infinite 0.0136 surface S5 Aspheric 19.8218 0.4821 Resin 1.54 56 APL5014CL
surface S6 Aspheric -2.5251 0.7337 surface S7 Aspheric -10.0889 1.3051 Resin 1.54 56 APL5014CL surface S8 Aspheric -1.5229 0.0760 surface
S9 Aspheric -1.4148 0.2782 Resin 1.65 22.9 EP7000 surface S10 Aspheric -1.9785 0.0500 surface S11 Extended -2.2632 0.7622 Resin 1.65 22.9
EP7000 aspheric surface S12 Extended 0.8529 1.0613 aspheric surface S13 Flat Infinite 0.2100 Glass 1.52 54.5 D263T surface Flat Infinite 0.0500

surface S14 Flat Infinite 0.0000 surface

(90) Table 3b gives conic coefficients k and polynomial coefficients a1, a2, a3, a4, a5, a6, a7, a8, a9, a10, a11, a12, a13, a14, and a15 that can be used for aspheric lenses surfaces S1 to S10 in Embodiment 3.

(91) TABLE-US-00008 TABLE 3b Parameter S1 S2 S3 S4 S5 S6 S7 S8 S9 S10 K -19.1982 -6.7071 -10.5697 -94.8320 -92.1353 -1.8945 39.5657 -0.6213 -0.6096 -0.5025 a1 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 a2 0.0956 -0.9439 -0.1502 0.4913 0.0435 0.1196 -0.1316 -0.3738 -0.6654 -0.5632 a3 0.1720 0.6845 0.5477 0.2394 0.0128 0.0550 -0.0684 0.1254 0.8416 0.6787 a4 0.0986 0.1394 -0.5466 -0.6546 -0.0759 -0.5139 0.2599 0.3174 0.5371 -0.0965 a5 -0.1024 -0.0329 0.8873 0.4402 0.0738 0.4193 -0.3706 0.5159 -0.4902 -0.4410 a6 -0.0022 -0.5623 -0.1384 2.7731 0.0413 -0.6119 -0.1841 0.0359 -0.1847 -0.0280 a7 -0.0098 0.1792 -0.9765 0.4146 -0.4024 0.2886 0.2694 -0.3266 0.0420 0.0990 a8 -0.0333 0.3229 0.6770 -9.6616 -0.2291 0.4303 0.6533 -0.3749 -0.0774 0.2199 a9 0.0279 -0.2649 -1.2748 -1.5490 0.6105 -1.1976 0.1466 -0.2612 0.0200 0.0806 a10 0.0284 0.2324 -1.1899 32.7032 -2.3182 -0.8407 -0.6587 -0.0882 0.0473 -0.0266 a11 -0.0325 -0.1094 1.5365 -15.6447 3.9257 -3.4558 -0.2067 0.0396 -0.0928 0.0011 a12 -0.0176 -0.1524 2.5773 -61.9447 7.7852 -5.5869 -0.1438 0.0833 -0.0936 -0.0157 a13 0.0005 0.1188 1.7996 17.9166 -0.9105 15.6206 0.1141 0.0908 -0.0681 -0.0193 a14 0.0203 0.0183 -0.9110 88.2484 7.6825 24.6383 0.3079 0.0386 -0.0172 -0.0141 a15 0.0196 0.0024 -2.7400 131.1438 -159.2654 2.0639 0.4387 0.0331 0.0679 -0.0040 a16 -0.0192 -0.0171 -1.9357 -227.7872 166.8148 -66.9990 -0.4919 0.0731 0.1369 0.0094

(92) Table 3c gives conic coefficients k and higher-order term coefficients X2Y0, X0Y2, X4Y0, X2Y2, X0Y4, X6Y0, X4Y2, X2Y4, X0Y6, X8Y0, X6Y2, X4Y4, X2Y6, and X0Y8 that can be used for the free-form surfaces S11 and S12 in Embodiment 3.

(93) TABLE-US-00009 TABLE 3c Parameter S11 S12 K -0.7265 -2.3120 X2Y0 1.2459 -0.4049 X0Y2 1.2611 -0.3527 X4Y0 -1.2282 -0.1014 X2Y2 -2.5043 -0.4262 X0Y4 -1.1911 -0.1977 X6Y0 0.9985 0.0226 X4Y2 2.8791 0.3820 X2Y4 2.6747 0.4401 X0Y6 0.7212 0.1124 X8Y0 -0.5561 0.0280 X6Y2 -1.7777 -0.1859 X4Y4 -1.6110 -0.1588 X2Y6 -0.9485 -0.1983 X0Y8 -0.0621 -0.0308

(94) FIG. 4C shows a distortion curve of the optical system in Embodiment 3, which represents distortion values corresponding to different fields of view.

(95) FIG. 4d shows a lateral chromatic aberration curve of the optical system in Embodiment 3, which represents lateral chromatic aberration values corresponding to five different wavelengths of light at different fields of view. Arrow indication lines are used to represent the five different wavelengths of light. The wavelengths are 510 nanometers, 470 nanometers, 610 nanometers, 550 nanometers, and 650 nanometers, respectively.

(96) It can be learned from FIG. 4c and FIG. 4d that the optical system provided in Embodiment 3 can achieve good imaging quality.

(97) Example embodiments of this application are described above. It should be noted that a person of ordinary skill in the art may further make several improvements and modifications without departing from the principle of this application, and these improvements and modifications are also considered to be within the protection scope of this application.

Claims

1. A camera module, comprising: a plurality of lenses, wherein lenses of the plurality of lenses are sequentially arranged from an object side to an image side along a direction of an optical axis, wherein at least one of the plurality of lenses is a free-form lens, wherein the free-form lens is a non-rotationally symmetric lens, wherein a first lens of the plurality of lenses is a lens of the plurality of lenses nearest the object side in a direction from the object side to the image side, wherein a distance on the optical axis between an object-side surface of the first lens and an imaging surface is TTL, wherein an effective focal length of the camera module is EFL, wherein $TTL/EFL \leq 2.0$, and wherein a lateral chromatic aberration of the camera module is less than 3 μm and a distortion is less than 2%; wherein one or more of an object-side surface or an image-side surface of the free-form lens are free-form surfaces, and wherein a surface type expression of the free-form lens is:

$$z = \frac{c_x x^2 + c_y y^2}{1 + \sqrt{1 - (1 + k_x) c_x^2 x^2 - (1 + k_y) c_y^2 y^2}} + \sum_{i=1}^M \text{Math. } A_i \cdot \text{Math. } x^i \cdot \text{Math. } + \sum_{i=1}^M \text{Math. } B_i \cdot \text{Math. } y^i \cdot \text{Math.}$$

wherein z is a sag of an optical surface, wherein x is an X-axis coordinate, and y is a Y-axis coordinate, wherein k.sub.x and k.sub.y are conic coefficients, wherein c.sub.x and c.sub.y are curvature radii, and wherein A.sub.i and B.sub.i are polynomial coefficients.

2. The camera module according to claim 1, wherein the free-form lens is symmetric with respect to a first plane, and wherein the free-form lens is also symmetric with respect to a second plane; and wherein the first plane is a plane comprising an X-axis and the optical axis, wherein the second plane is a plane comprising a Y-axis and the optical axis, and wherein the X-axis and the Y-axis are two central axes that are perpendicular to each other on the imaging surface of the camera module.

3. The camera module according to claim 2, wherein the X-axis and the Y-axis are each central axes passing through a center of the imaging surface and are respectively parallel to a long side and a short side of the imaging surface.

4. The camera module according to claim 3, wherein a quantity of lenses of the plurality of lenses is N, wherein $N \geq 3$, and wherein the plurality of lenses comprise the first lens to an Nth lens sequentially arranged in the direction from the object side to the image side; and wherein surfaces of object-side surfaces and image-side surfaces of the first lens to an (N-1)th lens are all aspheric surfaces, and wherein the Nth lens is a free-form lens.

5. The camera module according to claim 2, wherein a half of a diagonal length of an effective pixel region on the imaging surface of the camera module is ImgH, and wherein $TTL/ImgH \leq 2.0$.

6. The camera module according to claim 2, wherein an entrance pupil diameter of the camera module is EPD, and wherein $EFL/EPD \leq 2.2$.

7. The camera module according to claim 2, wherein a field of view of the camera module is FOV, wherein $FOV > 100$ degrees, and wherein $EFL < 20$ mm.

8. The camera module according to claim 1, wherein a quantity of the plurality of lenses is N, wherein three lenses that are of the plurality of lenses and that are nearest the object side are arranged sequentially in the direction from the object side to the image side and are, respectively, the first lens, a second lens, and a third lens; and wherein the camera module further comprises a vignetting stop, and wherein the vignetting stop is disposed on an object side of the second lens or on an object side of the third lens.

9. The camera module according to claim 1, wherein a lens of the plurality of lenses adjacent to the imaging surface is the free-form lens, wherein a curvature radius of the object-side surface of the free-form lens is R61, wherein a curvature radius of an image-side surface of the free-form lens is R62, and wherein $|EFL/R61| + |EFL/R62| < 2$.

10. The camera module according to claim 1, further comprising an electronic image sensor, wherein the electronic image sensor is disposed on the imaging surface, and wherein the imaging surface of the camera module is a rectangular region that matches an image sensing area of the electronic image sensor and is not less than the image sensing area of the electronic image sensor.

11. The camera module according to claim 10, wherein a diagonal length of an image sensing surface of the electronic image sensor is not less than 5.5 mm.

12. A terminal device, comprising: a camera module, comprising a plurality of lenses having lenses sequentially arranged from an object side to an image side along a direction of an optical axis, wherein at least one lens of the plurality of lenses is a free-form lens, wherein the free-form lens is a non-rotationally symmetric lens, wherein a first lens of the plurality of lenses is a lens of the plurality of lenses nearest the object side in a direction from the object side to the image side, wherein a distance on the optical axis between an object-side surface of the first lens and an imaging surface is TTL, wherein an effective focal length of the camera module is EFL, wherein $TTL/EFL \leq 2.0$, and wherein a lateral chromatic aberration of the camera module is less than 3 μm and a distortion is less than 2%; wherein one or more of an object-side surface or an image-side surface of the free-form lens are free-form surfaces, and wherein a surface type expression of the free-form lens is:

$$z = \frac{c_x x^2 + c_y y^2}{1 + \sqrt{1 - (1 + k_x)c_x^2 x^2 - (1 + k_y)c_y^2 y^2}} + \sum_{i=1}^M A_i \cdot \text{Math. } x^i \cdot \text{Math. } + \sum_{i=1}^M B_i \cdot \text{Math. } y^i \cdot \text{Math.}$$
 wherein z is a sag of an optical surface, wherein x is an X-axis coordinate, and y is a Y-axis coordinate, wherein k.sub.x and k.sub.y are conic coefficients, wherein c.sub.x and c.sub.y are curvature radii, and wherein A.sub.i and B.sub.i are polynomial coefficients.

13. The terminal device according to claim 12, wherein the free-form lens is symmetric with respect to a first plane, and wherein the free-form lens is symmetric with respect to a second plane; and wherein the first plane is a plane comprising an X-axis and the optical axis, wherein the second plane is a plane comprising a Y-axis and the optical axis, and wherein the X-axis and the Y-axis are each central axes that are perpendicular to each other on the imaging surface of the camera module.

14. A device, comprising: a plurality of lenses having lenses, including a first lens and a free-form lens, sequentially arranged from an object side to an image side along a direction of an optical axis; wherein the free-form lens is spaced apart from the first lens, wherein the first lens of the plurality of lenses is a lens of the plurality of lenses nearest the object side in a direction from the object side to the image side, wherein a distance on the optical axis between an object-side surface of the first lens and an imaging surface is TTL, wherein an effective focal length of the device is EFL, and wherein $TTL/EFL \leq 2.0$, and wherein a lateral chromatic aberration of the device is less than 3 μm and a distortion is less than 2%; and wherein one or more of an object-side surface or an image-side surface of the free-form lens are free-form surfaces, and wherein a surface type expression of the free-form lens is:

$$z = \frac{c_x x^2 + c_y y^2}{1 + \sqrt{1 - (1 + k_x)c_x^2 x^2 - (1 + k_y)c_y^2 y^2}} + \sum_{i=1}^M A_i \cdot \text{Math. } x^i \cdot \text{Math. } + \sum_{i=1}^M B_i \cdot \text{Math. } y^i \cdot \text{Math.}$$
 wherein z is a sag of an optical surface, wherein x is an X-axis coordinate, and y is a Y-axis coordinate, wherein k.sub.x and k.sub.y are conic coefficients, wherein c.sub.x and c.sub.y are curvature radii, and wherein A.sub.i and B.sub.i are polynomial coefficients.

15. The device according to claim 14, wherein the free-form lens is non-rotationally symmetric and is symmetric with respect to a first plane and symmetric with respect to a second plane different from the first plane; and wherein the first plane is a plane comprising an X-axis and the optical axis, wherein the second plane is a plane comprising a Y-axis and the optical axis, and wherein the X-axis and the Y-axis are two central axes that are perpendicular to each other on the imaging surface.

16. The device according to claim 15, wherein a quantity of lenses of the plurality of lenses is N, wherein $N \geq 3$, and wherein the plurality of lenses comprise the first lens to an Nth lens sequentially arranged in the direction from the object side to the image side; and wherein surfaces of object-side surfaces and image-side surfaces of the first lens to an (N-1)th lens are all aspheric surfaces, and wherein the Nth lens is the free-form lens.

17. The device according to claim 15, wherein the X-axis and the Y-axis are each central axes passing through a center of the imaging surface and are respectively parallel to a long side and a short side of the imaging surface.

18. The device according to claim 14, wherein a half of a diagonal length of an effective pixel region on the imaging surface of the device is ImgH, and wherein $TTL/\text{ImgH} \leq 2.0$.

19. The device according to claim 14, wherein a lens of the plurality of lenses adjacent to the imaging surface is the free-form lens, wherein a curvature radius of the object-side surface of the free-form lens is R61, wherein a curvature radius of an image-side surface of the free-form lens is R62, and wherein $|EFL/R61| + |EFL/R62| < 2$.
